

KATSUURA, Japan

WMO: 476740

Lat: 35.1508N

Lon: 140.3122E

Elev: 44

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.9	34.5	10.6	9.5	39.3	13.6	11.0	39.5	25.2	53.9	21.3	51.5	7.6	340	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.5	85.7	78.9	84.3	78.3	82.9	77.5	80.0	83.9	79.1	82.8	78.3	81.8	8.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.9	150.0	82.9	78.0	145.8	82.1	77.2	141.9	81.3	43.7	84.1	42.8	83.1	41.9	82.0	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	18.8	15.8	DB	29.3	89.0	2.2	1.5	27.7	90.1	26.5	91.0	25.3	91.8	23.7	93.0	(4)
(5)			WB	24.7	81.2	1.9	1.0	23.4	81.9	22.3	82.5	21.2	83.1	19.8	83.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.1	44.3	44.8	49.9	57.2	64.5	69.6	75.5	79.1	74.7	66.4	57.5	49.0
	DBStd	12.64	4.39	4.96	5.62	5.10	3.86	3.22	3.50	3.00	4.52	4.57	5.10	5.09
	HDD50	514	187	162	73	8	0	0	0	0	0	0	3	81
	HDD65	2733	642	567	467	235	55	5	0	0	1	36	228	497
	CDD50	4583	10	15	71	225	448	589	792	903	741	510	229	50
	CDD65	1325	0	0	0	1	38	144	327	438	292	81	4	0
	CDH74	7670	0	0	0	0	11	153	1750	3765	1806	185	0	0
	CDH80	1222	0	0	0	0	0	4	205	755	255	3	0	0
Wind	WSAvg	7.4	7.9	8.1	8.4	8.3	7.5	6.4	6.3	6.5	7.0	7.4	7.4	7.8
Precipitation	PrecAvg	74.6	3.8	3.9	6.4	6.0	6.5	8.1	6.1	4.5	9.3	10.0	6.1	3.4
	PrecMax	91.1	8.9	8.4	10.8	11.8	11.7	16.6	17.2	12.6	19.3	29.6	17.1	10.2
	PrecMin	50.9	0.1	0.6	0.9	2.7	3.4	3.1	0.2	0.0	3.0	1.5	1.2	0.2
	PrecStd	11.7	2.3	2.0	2.2	2.2	2.0	3.3	4.5	3.6	4.3	6.3	3.8	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.4	64.3	65.7	69.8	75.5	79.7	86.4	87.9	86.5	80.0	72.6	68.6
		MCWB	57.2	59.6	59.2	62.2	67.7	71.5	78.2	79.8	78.5	73.1	65.8	64.4
	2%	DB	59.4	61.0	63.7	67.9	73.4	77.0	84.0	86.3	84.4	77.9	70.7	64.9
		MCWB	53.4	56.2	58.0	62.3	66.4	71.1	77.8	79.4	78.0	72.3	64.3	58.8
	5%	DB	56.1	58.3	62.2	66.6	71.8	75.7	82.3	85.1	82.8	76.2	69.1	61.8
		MCWB	48.5	52.4	57.0	61.2	65.7	70.6	77.0	78.9	77.0	70.5	62.8	54.6
	10%	DB	53.0	55.0	60.5	65.3	70.5	74.5	80.7	83.8	81.3	74.3	67.0	58.3
		MCWB	45.0	47.8	54.9	60.1	65.1	70.0	76.3	78.2	76.2	68.2	60.6	50.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.1	60.8	62.1	66.5	70.4	74.5	80.0	81.5	80.1	75.7	68.5	65.6
		MCDB	61.8	63.3	64.0	67.9	72.8	77.0	84.0	85.8	84.0	78.1	71.0	68.1
	2%	WB	54.3	57.3	59.9	64.4	69.0	73.0	78.9	80.5	79.0	73.9	66.2	60.1
		MCDB	58.4	60.3	62.7	66.8	71.5	75.1	82.2	84.6	82.7	77.0	69.5	64.0
	5%	WB	49.4	53.1	58.0	62.7	67.8	72.0	77.9	79.8	78.1	71.7	63.9	55.6
		MCDB	54.7	57.8	61.5	65.5	70.4	74.4	81.1	83.6	81.7	75.1	68.0	60.8
	10%	WB	46.1	48.8	55.6	61.2	66.5	71.1	76.9	79.0	77.1	69.4	61.5	51.8
		MCDB	51.8	53.8	59.9	64.5	69.5	73.6	79.9	82.7	80.6	73.3	65.9	57.0
Mean Daily Temperature Range	5% DB	MDBR	13.6	13.4	12.9	11.6	10.0	8.3	7.8	8.5	9.3	10.6	12.4	13.1
		MCDBR	16.4	16.6	13.2	11.6	11.2	9.7	9.5	9.5	9.8	11.6	13.6	16.0
		MCWBR	14.1	15.4	11.9	9.0	7.3	5.7	5.0	4.8	5.5	8.3	10.9	13.4
	5% WB	MCDBR	15.8	16.1	12.6	9.4	9.1	7.6	8.7	8.7	9.0	10.5	12.7	15.5
		MCWBR	14.7	15.7	12.9	8.8	7.3	5.3	4.9	4.8	5.7	9.3	11.7	14.5
Clear-Sky Solar Irradiance	taub	0.388	0.440	0.504	0.547	0.558	0.571	0.568	0.546	0.495	0.479	0.432	0.390	
	taud	2.229	2.081	1.930	1.873	1.907	1.948	2.002	2.055	2.172	2.132	2.174	2.234	
	Ebn at Noon	250	249	243	239	239	235	235	237	244	234	232	241	
	Edh at Noon	35	45	57	63	62	59	56	52	44	43	36	33	
All-Sky Solar Radiation	RadAvg	822	961	1238	1474	1610	1429	1638	1672	1269	941	759	726	
	RadStd	67	110	91	152	166	140	269	142	108	95	69	61	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+0.67	+0.83	+0.97	N/A	N/A	+169	+117
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page